# Vector Inequalities and Theorems

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21 January, 2026

## 1 Cauchy–Schwarz Inequality

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ be two vectors. The **Cauchy–Schwarz inequality** states that

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|.$$

Here, $\mathbf{u} \cdot \mathbf{v}$ denotes the dot product of the vectors, and

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}$$

is the magnitude (or norm) of $\mathbf{u}$.

**Geometric interpretation:** Since

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta,$$

where $\theta$ is the angle between the vectors, the inequality follows from the fact that

$$|\cos\theta| \leq 1.$$

Equality holds if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

### 1.1 Verification of the Cauchy–Schwarz Inequality

Let $\mathbf{u} = (3, 4)$ and $\mathbf{v} = (2, -3)$.

First, compute the dot product:

$$\mathbf{u} \cdot \mathbf{v} = (3)(2) + (4)(-3) = 6 - 12 = -6.$$

Thus,

$$|\mathbf{u} \cdot \mathbf{v}| = 6.$$

Next, compute the magnitudes of the vectors:

$$\|\mathbf{u}\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5,$$

$$\|\mathbf{v}\| = \sqrt{2^2 + (-3)^2} = \sqrt{13}.$$

Now, compute the product of the magnitudes:

$$\|\mathbf{u}\|\,\|\mathbf{v}\| = 5\sqrt{13}.$$

Since

$$6 \leq 5\sqrt{13},$$

the Cauchy–Schwarz inequality

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\,\|\mathbf{v}\|$$

is verified for the given vectors.

# 2 Triangle Inequality

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. The **triangle inequality** for vectors states that

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

**Explanation:** This inequality expresses the fact that the length of one side of a triangle is always less than or equal to the sum of the lengths of the other two sides. When vectors are placed head-to-tail, $\mathbf{u} + \mathbf{v}$ represents the third side of the triangle.

Equality occurs if and only if $\mathbf{u}$ and $\mathbf{v}$ point in the same direction.

## 2.1 Verification of the Triangle Inequality

Let $\mathbf{u} = (1, 1, 1)$ and $\mathbf{v} = (0, 1, -2)$.

First, compute the sum of the vectors:

$$\mathbf{u} + \mathbf{v} = (1 + 0,\ 1 + 1,\ 1 - 2) = (1, 2, -1).$$

Next, compute the magnitudes of the vectors:

$$\|\mathbf{u}\| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3},$$

$$\|\mathbf{v}\| = \sqrt{0^2 + 1^2 + (-2)^2} = \sqrt{5}.$$

Now, compute the magnitude of the sum:

$$\|\mathbf{u} + \mathbf{v}\| = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6}.$$

Finally, compute the sum of the magnitudes:

$$\|\mathbf{u}\| + \|\mathbf{v}\| = \sqrt{3} + \sqrt{5}.$$

Since

$$\sqrt{6} \leq \sqrt{3} + \sqrt{5},$$

the triangle inequality

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$$

is verified for the given vectors.

# 3 Pythagorean Theorem

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ be vectors such that

$$\mathbf{u} \cdot \mathbf{v} = 0.$$

This means that $\mathbf{u}$ and $\mathbf{v}$ are orthogonal (perpendicular).

The **Pythagorean theorem for vectors** states that

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

**Proof:**

$$\|\mathbf{u} + \mathbf{v}\|^2 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = \mathbf{u} \cdot \mathbf{u} + 2\,\mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{v}.$$

Since $\mathbf{u} \cdot \mathbf{v} = 0$, this simplifies to

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

This result generalizes the classical Pythagorean theorem to vector spaces.

## 3.1 Verification of the Pythagorean Theorem

Let $\mathbf{u} = (3, 4, -2)$ and $\mathbf{v} = (4, -3, 0)$.

First, compute the dot product:

$$\mathbf{u} \cdot \mathbf{v} = (3)(4) + (4)(-3) + (-2)(0) = 12 - 12 + 0 = 0.$$

Since $\mathbf{u} \cdot \mathbf{v} = 0$, the vectors are orthogonal.

Next, compute the magnitudes of the vectors:

$$\|\mathbf{u}\|^2 = 3^2 + 4^2 + (-2)^2 = 9 + 16 + 4 = 29,$$

$$\|\mathbf{v}\|^2 = 4^2 + (-3)^2 + 0^2 = 16 + 9 + 0 = 25.$$

Now, compute the sum of the vectors:

$$\mathbf{u} + \mathbf{v} = (7, 1, -2).$$

Then, compute the square of the magnitude of the sum:

$$\|\mathbf{u} + \mathbf{v}\|^2 = 7^2 + 1^2 + (-2)^2 = 49 + 1 + 4 = 54.$$

Since

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 = 29 + 25 = 54,$$

the Pythagorean theorem

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

is verified for the given vectors.

## 4 Verification of the Statement

1)The given statement is

$$\|\mathbf{v}\| = |v_1 + v_2 + \cdots + v_n|.$$

This statement is **false**.
**Counterexample:** Let $\mathbf{v} = (1, -1, 0, \ldots, 0) \in \mathbb{R}^n$.
Then,

$$\|\mathbf{v}\| = \sqrt{1^2 + (-1)^2} = \sqrt{2},$$

while

$$|v_1 + v_2 + \cdots + v_n| = |1 - 1 + 0 + \cdots + 0| = 0.$$

Since

$$\sqrt{2} \neq 0,$$

the given statement is false.
2)The given statement is:

$$\text{If } \mathbf{u} \cdot \mathbf{v} < 0, \text{ then the angle between } \mathbf{u} \text{ and } \mathbf{v} \text{ is acute.}$$

This statement is **false**.
**Reason / Counterexample:** For any two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$,

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta,$$

where $\theta$ is the angle between the vectors.
If $\mathbf{u} \cdot \mathbf{v} < 0$, then

$$\cos\theta < 0,$$

which implies that

$$\theta > 90^\circ.$$

Hence, the angle between $\mathbf{u}$ and $\mathbf{v}$ is **obtuse**, not acute.
Therefore, the given statement is false.

## 5 Meaningless expression

1) Explain why the expression $\|u \cdot v\|$ is meaningless.

The expression $\|u \cdot v\|$ is meaningless because $u \cdot v$ denotes the dot product of vectors $u$ and $v$, which is a scalar, whereas the norm $\| \cdot \|$ is defined for vectors, not for scalars.

2) Explain why the expression $u + (u \cdot v)$ is meaningless.

The expression $u + (u \cdot v)$ is meaningless because $u$ is a vector while $u \cdot v$ is a scalar, and vector addition between a vector and a scalar is not defined.

## 6 Wronskian and Linear Independence

The Wronskian of a set of $n$ functions $f_1, f_2, \ldots, f_n$, which are $(n-1)$ times differentiable on an interval $I$, is defined as the determinant

$$W(f_1, f_2, \ldots, f_n)(x) = \begin{vmatrix} f_1(x) & f_2(x) & \cdots & f_n(x) \\ f_1'(x) & f_2'(x) & \cdots & f_n'(x) \\ f_1''(x) & f_2''(x) & \cdots & f_n''(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x) \end{vmatrix}.$$

The Wronskian is used to study the linear independence of functions. If there exists a point $x_0 \in I$ such that $W(x_0) \neq 0$, then the functions $f_1, f_2, \ldots, f_n$ are linearly independent on the interval $I$. If $W(x) = 0$ for all $x \in I$, and the functions are solutions to a homogeneous linear differential equation, they are linearly dependent.

Listing 1: Wronskian computation using SymPy

```
import sympy
import numpy as np
import matplotlib.pyplot as plt

def wronskian(functions, x):
    """
    Calculates the Wronskian matrix and its determinant for a list
       of functions
    with respect to a variable x.

    Args:
        functions (list): A list of sympy symbolic functions.
        x (sympy.Symbol): The symbolic variable with respect to
           which to differentiate.

    Returns:
        tuple: A tuple containing the Wronskian matrix
           (sympy.Matrix) and
               its determinant (sympy.Expr).
    """
    n = len(functions)
    if n == 0:
        return sympy.Matrix([]), 0

    # Create the Wronskian matrix
    # Each row corresponds to a derivative order, each column to a
       function.
    wronskian_matrix = sympy.Matrix.zeros(n, n)
    # Loop for derivative order (rows)
    for j in range(n):
```

```
        # Loop for functions (columns)
        for i in range(n):
            current_func = functions[i]
            # Calculate the j-th derivative of the i-th function
            derivative = sympy.diff(current_func, x, j)
            wronskian_matrix[j, i] = derivative

    # Calculate the determinant of the Wronskian matrix
    determinant = wronskian_matrix.det()
    return wronskian_matrix, determinant

# --- Interactive Example Usage ---

x = sympy.symbols('x')

print("Enter functions as symbolic expressions (e.g., 'sin(x)',
   'x**2', 'exp(x)').")

input_functions_str = []

while True:
    try:
        num_functions = int(input("How many functions do you want
           to enter? "))
        if num_functions < 0:
            print("Please enter a non-negative number.")
        else:
            break
    except ValueError:
        print("Invalid input. Please enter an integer.")

for i in range(num_functions):
    func_str = input(f"Enter function {i + 1}: ")
    input_functions_str.append(func_str)

if not input_functions_str:
    print("No functions entered. Exiting.")
else:
    try:
        # Convert string inputs to sympy symbolic functions,
           ensuring 'e' is recognized as sympy.E
        symbolic_functions = [sympy.sympify(f_str, locals={'x': x,
           'sin': sympy.sin, 'cos': sympy.cos, 'exp': sympy.exp,
           'log': sympy.log, 'e': sympy.E}) for f_str in
           input_functions_str]

        print(f"\nCalculating Wronskian for functions:
```

```
    {symbolic_functions} with respect to {x}")
wronskian_mat, wronskian_det =
    wronskian(symbolic_functions, x)

print("\n--- Wronskian Matrix ---")
display(wronskian_mat) # Using display for better matrix
    rendering in Colab

# Print the simplified determinant
print("\n--- Wronskian Determinant ---")
simplified_det = sympy.simplify(wronskian_det)
print(simplified_det)

# --- Plotting the Wronskian Determinant and individual
    functions ---
print("\n--- Plotting Wronskian Determinant and Input
    Functions ---")

# Convert the sympy expression to a numerical function
wronskian_func = sympy.lambdify(x, simplified_det, 'numpy')

# Define the range for x
x_vals = np.linspace(-5, 5, 400) # From -5 to 5 with 400
    points

# Evaluate the Wronskian function over the x range
y_vals_wronskian = wronskian_func(x_vals)

# Handle scalar output for constant Wronskian determinant
if np.isscalar(y_vals_wronskian):
    y_vals_wronskian = np.full_like(x_vals,
        y_vals_wronskian)

# Create the plot
plt.figure(figsize=(12, 8))
plt.plot(x_vals, y_vals_wronskian, label=f'Wronskian:
    {simplified_det}', linewidth=2, linestyle='--')

# Plot each individual function
for func_sym in symbolic_functions:
    func_lambdified = sympy.lambdify(x, func_sym, 'numpy')
    func_y_vals = func_lambdified(x_vals)

    # Handle scalar output for constant functions
    if np.isscalar(func_y_vals):
        func_y_vals = np.full_like(x_vals, func_y_vals)
```

```
            plt.plot(x_vals, func_y_vals, label=f'Function: {func_sym}')

        plt.title('Plot of Wronskian Determinant and Input Functions')
        plt.xlabel('x')
        plt.ylabel('Value')
        plt.grid(True)
        plt.axhline(0, color='black', linewidth=0.5)
        plt.axvline(0, color='black', linewidth=0.5)
        plt.legend()
        plt.show()

        # --- Analysis of Linear Independence ---
        print("\n--- Analysis of Linear Independence ---")
        # Check if the determinant is identically zero. sympy.Eq can compare symbolic expressions.
        # If simplified_det is a constant (like 0), sympy.Eq will handle it.
        if sympy.Eq(simplified_det, 0):
            print("The Wronskian determinant is identically zero for all x. This test is inconclusive; the functions may or may not be linearly independent.")
        else:
            # If the Wronskian is not identically zero, it means it's non-zero at at least one point.
            print("The Wronskian determinant is not identically zero. Therefore, the functions are linearly independent.")

    except (sympy.SympifyError, NameError) as e:
        print(f"Error parsing function: {e}. Please ensure correct symbolic syntax.")
```

## Problems

1)Find the Wronskian of the set of functions

$$\{1, x, e^x\}.$$

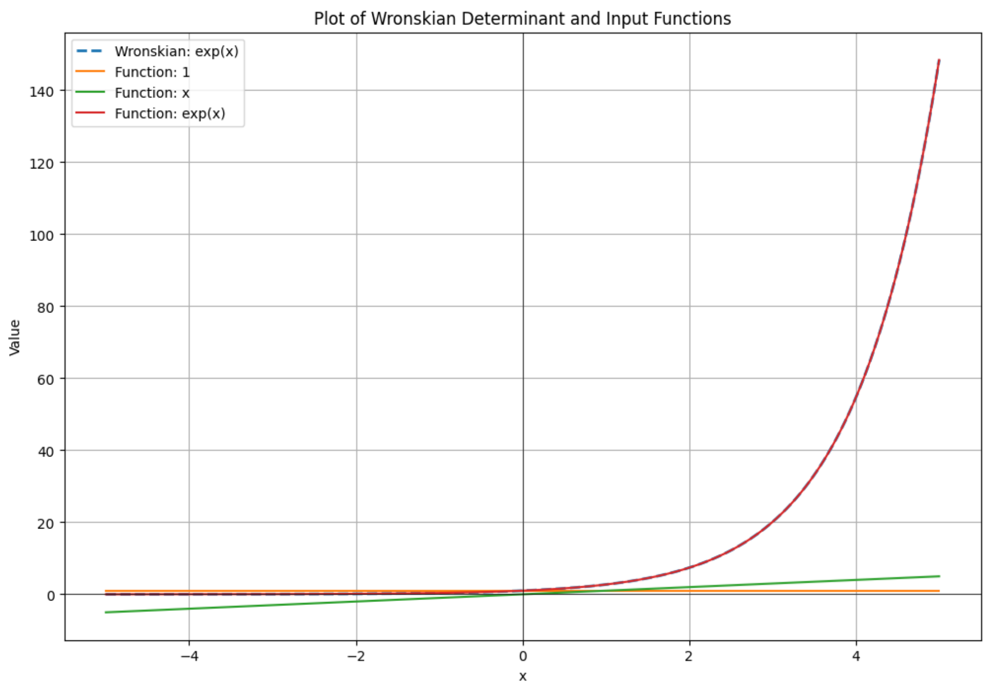
OUTPUT =>Enter functions as symbolic expressions (e.g., 'sin(x)', 'x**2', 'exp(x)').
How many functions do you want to enter? 3
Enter function 1:1
Enter function 2: x
Enter function 3: e**x

Calculating Wronskian for functions: [1, x, exp(x)] with respect to x
— Wronskian Matrix —

$$\begin{bmatrix} 1 & x & e^x \\ 0 & 1 & e^x \\ 0 & 0 & e^x \end{bmatrix}$$

— Wronskian Determinant — exp(x)

— Plotting Wronskian Determinant and Input Functions —




— Analysis of Linear Independence —

The Wronskian determinant is not identically zero. Therefore, the functions are **linearly independent**.

2)Find the Wronskian of the set of functions

{1, sin(2x), cos(2x)}.

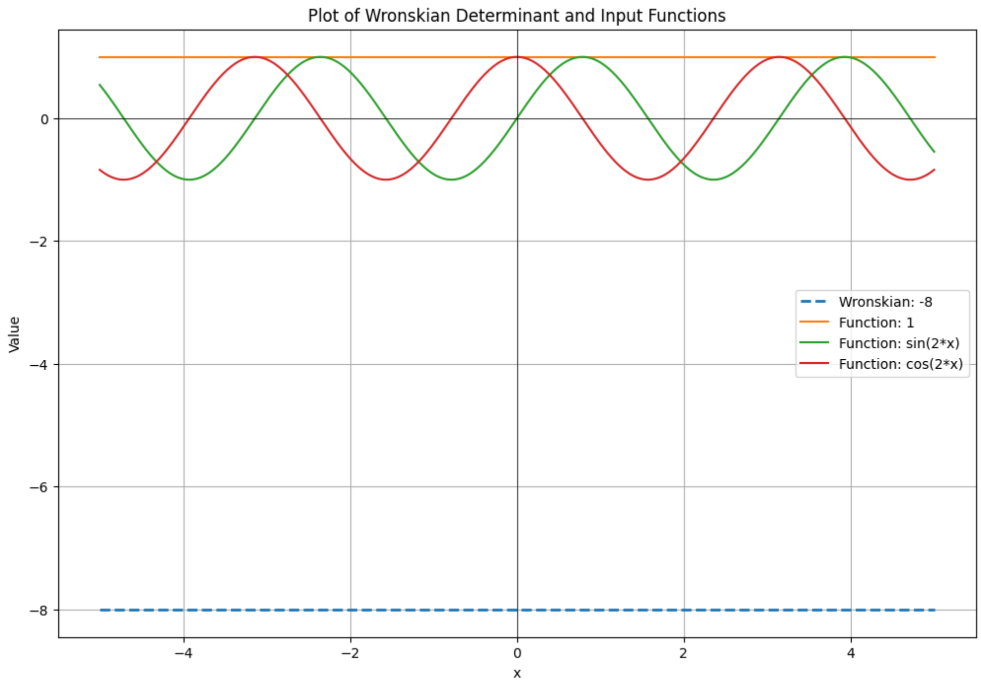
OUTPUT =>Enter functions as symbolic expressions (e.g., 'sin(x)', 'x**2', 'exp(x)').
How many functions do you want to enter? 3
Enter function 1:1
Enter function 2: sin(2*x)
Enter function 3: cos(2*x)
Calculating Wronskian for functions: [1, sin(2*x), cos(2*x)] with respect to x

— Wronskian Matrix —

$$\begin{bmatrix} 1 & sin(2x) & cos(2x) \\ 0 & 2cos(2x) & -2sin(2x) \\ 0 & -4sin(2x) & -4cos(2x) \end{bmatrix}$$

— Wronskian Determinant —
-8

—Plotting Wronskian Determinant and Input Functions —




— Analysis of Linear Independence —
The Wronskian determinant is not identically zero. Therefore, the functions are **linearly independent**.

3)Test the following set of solutions for linear independence:

$$y'' + 6y' + 9y = 0, \qquad \{e^{-3x}, xe^{-3x}\}$$

OUTPUT =>Enter functions as symbolic expressions (e.g., 'sin(x)', 'x**2', 'exp(x)').
How many functions do you want to enter? 2
Enter function 1: e**(-3*x)
Enter function 2: x*e**(-3*x)

Calculating Wronskian for functions: [exp(-3*x), x*exp(-3*x)] with respect to x
— Wronskian Matrix —

$$\begin{bmatrix} e^{-3x} & xe^{-3x} \\ -3e^{-3x} & -3xe^{-3x} + e^{-3x} \end{bmatrix}$$

— Wronskian Determinant —
exp(-6*x)
— Analysis of Linear Independence — The Wronskian determinant is not identically zero. Therefore, the functions are **linearly independent**.

4)Test the following set of solutions for linear independence:

$$y''' - 6y'' + 11y' - 6y = 0, \qquad \{e^x, e^{2x}, e^x - e^{2x}\}.$$

OUTPUT =>Enter functions as symbolic expressions (e.g., 'sin(x)', 'x**2', 'exp(x)').
How many functions do you want to enter? 3
Enter function 1: e**x
Enter function 2: e**(2*x)
Enter function 3: e**x-e**(2*x)

Calculating Wronskian for functions: [exp(x), exp(2*x), -exp(2*x) + exp(x)] with respect to x

— Wronskian Matrix —

$$\begin{bmatrix} e^x & e^{2x} & -e^{2x} + e^x \\ e^x & 2e^{2x} & -2e^{2x} + e^x \\ e^x & 4e^{2x} & (1 - 4e^x)e^x \end{bmatrix}$$

— Wronskian Determinant —
0

If the Wronskian of a set of functions is zero, linear dependence does not follow in general. However, for solutions of a linear homogeneous differential equation with continuous coefficients, an identically zero Wronskian implies linear dependence.Since these functions are given to be solutions of the differential equation (which can also be verified directly), the given set of solutions is linearly dependent.

$$y_1 = e^x, \quad y_2 = e^{2x}, \quad y_3 = e^x - e^{2x}$$

$$y''' - 6y'' + 11y' - 6y = 0$$

—

1) Check $y_1 = e^x$:

$$\begin{aligned} y_1' &= e^x \\ y_1'' &= e^x \\ y_1''' &= e^x \\ y_1''' - 6y_1'' + 11y_1' - 6y_1 &= e^x - 6e^x + 11e^x - 6e^x \\ &= (1 - 6 + 11 - 6)e^x = 0 \end{aligned}$$

2) Check $y_2 = e^{2x}$:

$$
\begin{aligned}
y_2' &= 2e^{2x} \\
y_2'' &= 4e^{2x} \\
y_2''' &= 8e^{2x} \\
y_2''' - 6y_2'' + 11y_2' - 6y_2 &= 8e^{2x} - 6(4e^{2x}) + 11(2e^{2x}) - 6e^{2x} \\
&= (8 - 24 + 22 - 6)e^{2x} = 0
\end{aligned}
$$

3) Check $y_3 = e^x - e^{2x}$:

$$
\begin{aligned}
y_3' &= e^x - 2e^{2x} \\
y_3'' &= e^x - 4e^{2x} \\
y_3''' &= e^x - 8e^{2x} \\
y_3''' - 6y_3'' + 11y_3' - 6y_3 &= (e^x - 8e^{2x}) - 6(e^x - 4e^{2x}) + 11(e^x - 2e^{2x}) - 6(e^x - e^{2x}) \\
&= (1 - 6 + 11 - 6)e^x + (-8 + 24 - 22 + 6)e^{2x} = 0
\end{aligned}
$$

Hence, all three functions satisfy the differential equation.
Since the Wronskian is identically zero and all functions satisfy the same linear homogeneous differential equation, the set

$$
\{e^x,\, e^{2x},\, e^x - e^{2x}\}
$$

is **linearly dependent**.

# Conclusion

This assignment explored several key concepts in vector analysis and linear differential equations through computational experiments and visualizations. We verified fundamental vector inequalities, such as the Cauchy-Schwarz, Pythagorean, and Triangle inequalities,and additionally, we tested sets of functions for linear independence using Wronskian computations. The results from code outputs and plots confirmed theoretical expectations, illustrating the connection between vector properties and linear dependence of functions.